OS & SP

Name : D.Teja Sadwika **Roll No :** 2520030536

SKILL – 09

1.Skill9.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#define MAX_INPUT 1024

char previous_directory[PATH_MAX] = "";

/* Display current directory */
void show_directory(void)
{
   char current[PATH_MAX];
```

```
    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return;
    }

    printf("%s\n", current);
}

/* Change directory */
void change_directory(char *directory)
{
    char current[PATH_MAX];

    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return;
    }
```

```
    if (chdir(directory) != 0) {
        perror("cd");
        return;
    }

    strcpy(previous_directory, current);
    printf("Directory changed successfully.\n");
}

/* Return to previous directory */
void go_previous(void)
{
    if (previous_directory[0] == '\0') {
        printf("No previous directory available.\n");
        return;
    }

    char current[PATH_MAX];
```

```
    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return;
    }

    if (chdir(previous_directory) != 0) {
        perror("prev");
        return;
    }

    strcpy(previous_directory, current);
    printf("Directory changed successfully.\n");
}

/* Display help */
void show_help(void)
{
```

```
    printf("\nAvailable commands:\n");
    printf("  pwd          - show current directory\n");
    printf("  cd <dir>     - change directory\n");
    printf("  cd           - change to home directory\n");
    printf("  prev         - return to previous directory\n");
    printf("  help         - display commands\n");
    printf("  exit         - terminate program\n\n");
}

int main(void)
{
    char input[MAX_INPUT];

    printf("========================================\n");
    printf("         OSSP PROGRAM 9\n");
    printf("         Directory Navigation\n");
    printf("========================================\n");
```

```
printf("\nType 'help' for commands.\n\n");

while (1) {
    printf("shell9> ");
    fflush(stdout);

    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (input[0] == '\0')
        continue;

    if (strcmp(input, "exit") == 0)
        break;

    if (strcmp(input, "help") == 0) {
```

```
    show_help();
}
else if (strcmp(input, "pwd") == 0) {
    show_directory();
}
else if (strcmp(input, "prev") == 0) {
    go_previous();
}
else if (strcmp(input, "cd") == 0) {
    char *home = getenv("HOME");

    if (home == NULL) {
        printf("HOME environment variable is not set.\n");
    } else {
        change_directory(home);
    }
}
else if (strncmp(input, "cd ", 3) == 0) {
```

```
        char *directory = input + 3;

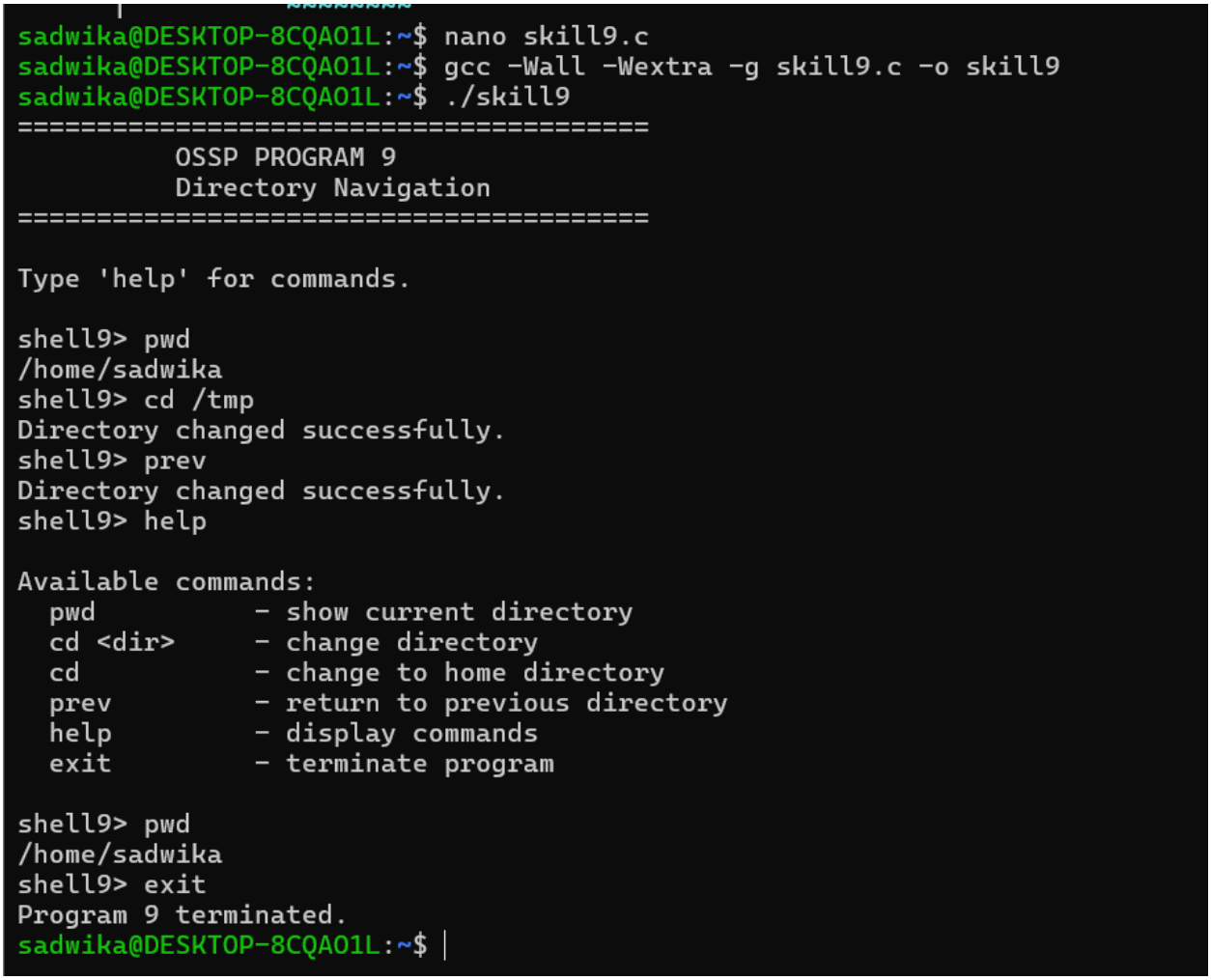
        while (*directory == ' ' || *directory == '\t')
            directory++;

        if (*directory == '\0') {
            printf("Usage: cd <directory>\n");
        } else {
            change_directory(directory);
        }
    }
    else {
        printf("Unknown command: %s\n", input);
        printf("Type 'help' for available commands.\n");
    }
}

printf("Program 9 terminated.\n");
```

```
    return 0;
}
```

OUTPUT :



```
sadwika@DESKTOP-8CQAO1L:~$ nano skill9.c
sadwika@DESKTOP-8CQAO1L:~$ gcc -Wall -Wextra -g skill9.c -o skill9
sadwika@DESKTOP-8CQAO1L:~$ ./skill9
========================================
          OSSP PROGRAM 9
          Directory Navigation
========================================

Type 'help' for commands.

shell9> pwd
/home/sadwika
shell9> cd /tmp
Directory changed successfully.
shell9> prev
Directory changed successfully.
shell9> help

Available commands:
  pwd           - show current directory
  cd <dir>      - change directory
  cd            - change to home directory
  prev          - return to previous directory
  help          - display commands
  exit          - terminate program

shell9> pwd
/home/sadwika
shell9> exit
Program 9 terminated.
sadwika@DESKTOP-8CQAO1L:~$
```